

LeafLens: Tomato Leaf Disease Detection with YOLOv8 on a Raspberry Pi 5

Shermaine Yap Shi Min

A project submitted to the School of Computing, Asia Pacific University

Declaration

I declare this is my own work.

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Abstract

Smallholder tomato farmers in Cameron Highlands lose up to 30% of yield to early blight, late blight and leaf mould. LeafLens is an offline edge system: a YOLOv8n detector on a Raspberry Pi 5 localises lesions and classifies six classes. On 4,212 images it reaches mAP@0.5 of 0.87 on a field test set and runs at 118 ms per image after INT8 quantisation with NCNN.

1. Introduction

Tomato is a key crop in the highlands. Diagnosis relies on monthly visits by extension officers. Tomato is a key crop in the highlands. Diagnosis relies on monthly visits by extension officers. Tomato is a key crop in the highlands. Diagnosis relies on monthly visits by extension officers. Tomato is a key crop in the highlands. Diagnosis relies on monthly visits by extension officers. Tomato is a key crop in the highlands. Diagnosis relies on monthly visits by extension officers.

2. Literature Review

Prior work on PlantVillage reports high accuracy on lab images but poor field transfer. Prior work on PlantVillage reports high accuracy on lab images but poor field transfer. Prior work on PlantVillage reports high accuracy on lab images but poor field transfer. Prior work on PlantVillage reports high accuracy on lab images but poor field transfer. Prior work on PlantVillage reports high accuracy on lab images but poor field transfer. Prior work on PlantVillage reports high accuracy on lab images but poor field transfer. Prior work on PlantVillage reports high accuracy on lab images but poor field transfer. Prior work on PlantVillage reports high accuracy on lab images but poor field transfer. Prior work on PlantVillage reports high accuracy on lab images but poor field transfer. Prior work on PlantVillage reports high accuracy on lab images but poor field transfer. Prior work on PlantVillage reports high accuracy on lab images but poor field transfer. Prior work on PlantVillage reports high accuracy on lab images but poor field transfer. Prior work on PlantVillage reports high accuracy on lab images but poor field transfer. Prior work on PlantVillage reports high accuracy on lab images but poor field transfer. Prior work on PlantVillage reports high accuracy on lab images but poor field transfer. Prior work on PlantVillage reports high accuracy on lab images but poor field transfer. Prior work on PlantVillage reports high accuracy on lab images but poor field transfer. Prior work on PlantVillage reports high accuracy on lab images but poor field transfer. Prior work on PlantVillage reports high accuracy on lab images but poor field transfer. Prior work on PlantVillage reports high accuracy on lab images but poor field transfer.

3. Methodology

We collected 1,120 field images across three farms and combined them with PlantVillage and PlantDoc, annotated in Roboflow with 9,870 boxes. Mosaic augmentation and background replacement reduce the lab-to-field gap. We collected 1,120 field images across three farms and combined them with PlantVillage and PlantDoc, annotated in Roboflow with 9,870 boxes. Mosaic augmentation and background replacement reduce the lab-to-field gap. We collected 1,120 field images across three farms and combined them with PlantVillage and PlantDoc, annotated in Roboflow with 9,870 boxes. Mosaic augmentation and background replacement reduce the lab-to-field gap. We collected 1,120 field images across three farms and combined them with PlantVillage and PlantDoc, annotated in Roboflow with 9,870 boxes. Mosaic augmentation and background replacement reduce the lab-to-field gap.

4. Results

Field mAP@0.5 is 0.87 versus 0.95 on PlantVillage. Late blight versus early blight is the main confusion. INT8 inference takes 118 ms on the Pi 5 versus 410 ms for YOLOv8s. A SUS study with 9 farmers scored 78. Field mAP@0.5 is 0.87 versus 0.95 on PlantVillage. Late blight versus early blight is the main confusion. INT8 inference takes 118 ms on the Pi 5 versus 410 ms for YOLOv8s. A SUS study with 9 farmers scored 78. Field mAP@0.5 is 0.87 versus 0.95 on PlantVillage. Late blight versus early blight is the main confusion. INT8 inference takes 118 ms on the Pi 5 versus 410 ms for YOLOv8s. A SUS study with 9 farmers scored 78.

5. Conclusion

LeafLens works offline at usable speed; limitations are daytime-only images and a single region. LeafLens works offline at usable speed; limitations are daytime-only images and a single region. LeafLens works offline at usable speed; limitations are daytime-only images and a single region.

References

[1] Redmon, J. You only look once. [2] Hughes, D. PlantVillage.